

2026 NESS Statathon:

Policy Retention Modeling for Travelers Insurance

Ronnie Orsini

Competition Overview

Travelers asked us to build a model that helps them retain customers. Three deliverables:

- Predict which policies will cancel
- Understand the drivers behind cancellation
- Recommend how Travelers can use the model to improve retention



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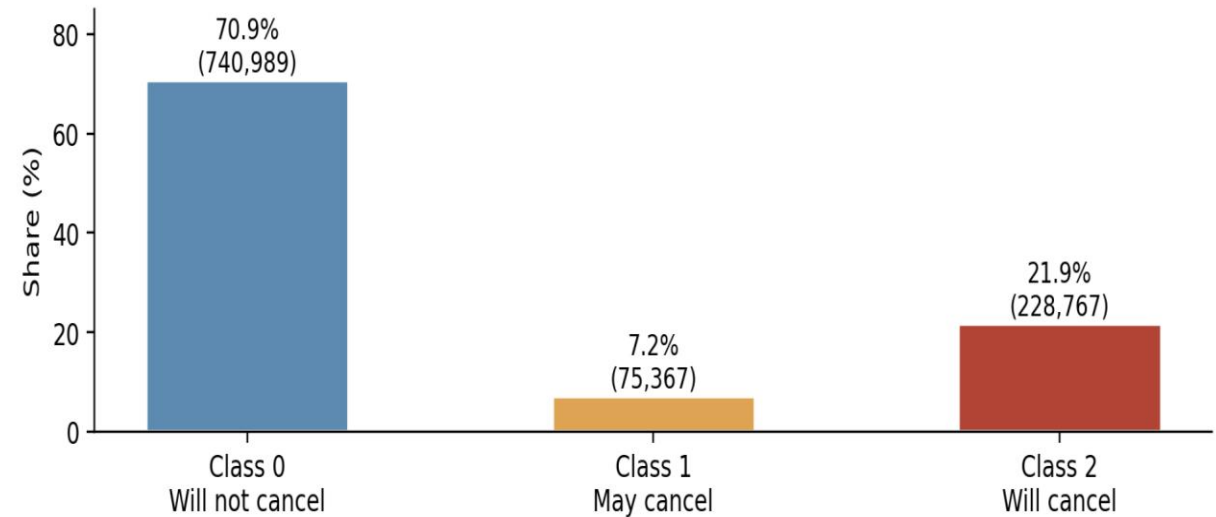
**New England
Statistical Society**

NEW ENGLAND STATISTICS SYMPOSIUM SINCE 1987 & NEW ENGLAND STATISTICAL SOCIETY SINCE 2017

Data Overview

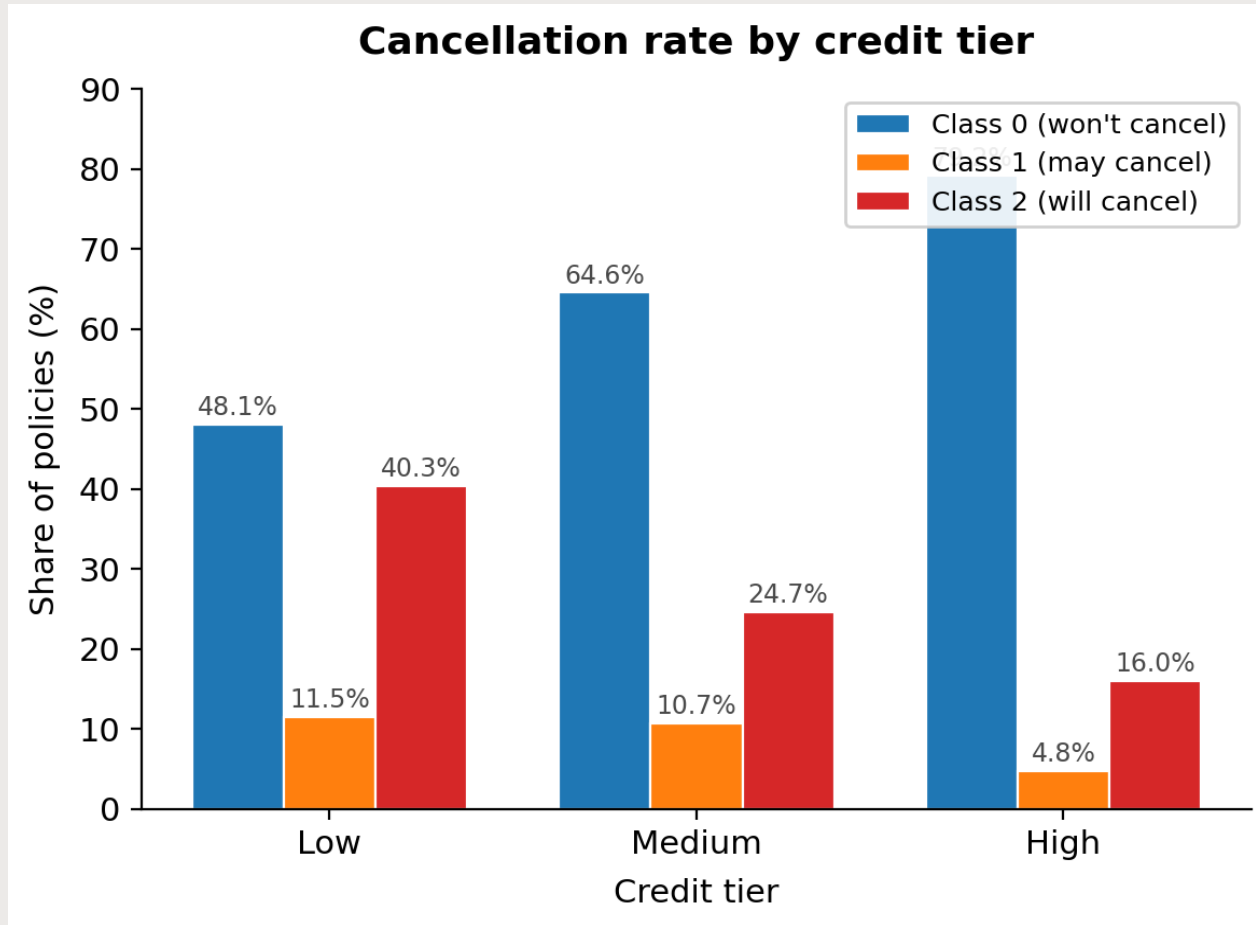
The data includes:

- **Policyholder info:** age, gender, marital status, credit
- **Policy info:** premium, tenure, coverage type, claim indicator
- **Property info:** dwelling type, square footage, house color
- **Timing info:** year, renewal season, quote weekday
- **Sales/geography:** sales channel, zip code



Each row represents a policy record for a given policy year/term 2013 through 2017.

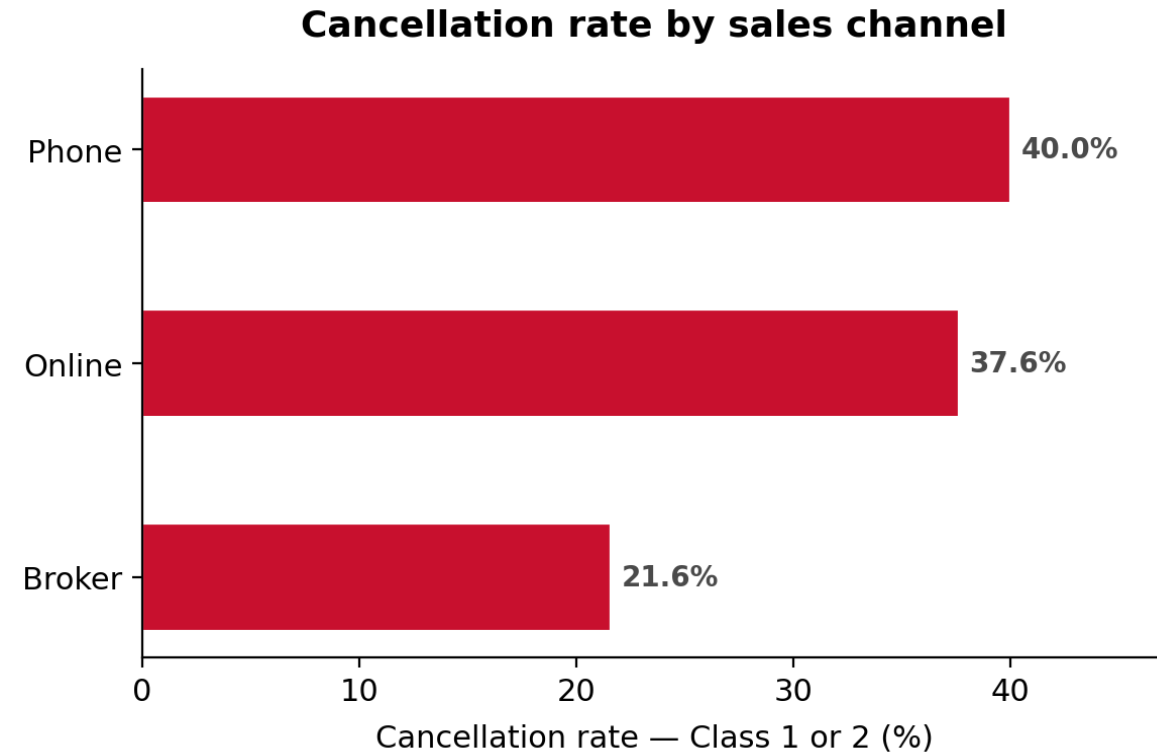
Exploratory Data Analysis



- Low credit policies cancelled at 51% combined, vs 22% for high credit
- Younger policyholders and rare zip codes showed elevated risk
 - Age interactions
 - OOF target encoding

Exploratory Data Analysis

- Phone and Online channels cancelled near 40%, vs 22% for Broker
- Premium, tenure, and dwelling type added smaller complementary signal
 - Ratios and interactions



21 raw predictors → Expanded model feature set

Raw Policy Variables

- 21 usable predictors
- Customer: age, credit, household info
- Policy: premium, tenure, coverage type
- Property: zip code, dwelling type, square footage
- Behavior: sales channel, renewal season, claim indicator

Feature Engineering

- Interactions: claim × tenure, tenure × credit
- Ratios: premium per sqft, premium vs zip/credit/dwelling
- Transforms: log premium, log tenure, log square footage
- Missingness flags: missing credit, age, premium, etc.
- Encoding: one hot, frequency/count, OOF target encoding

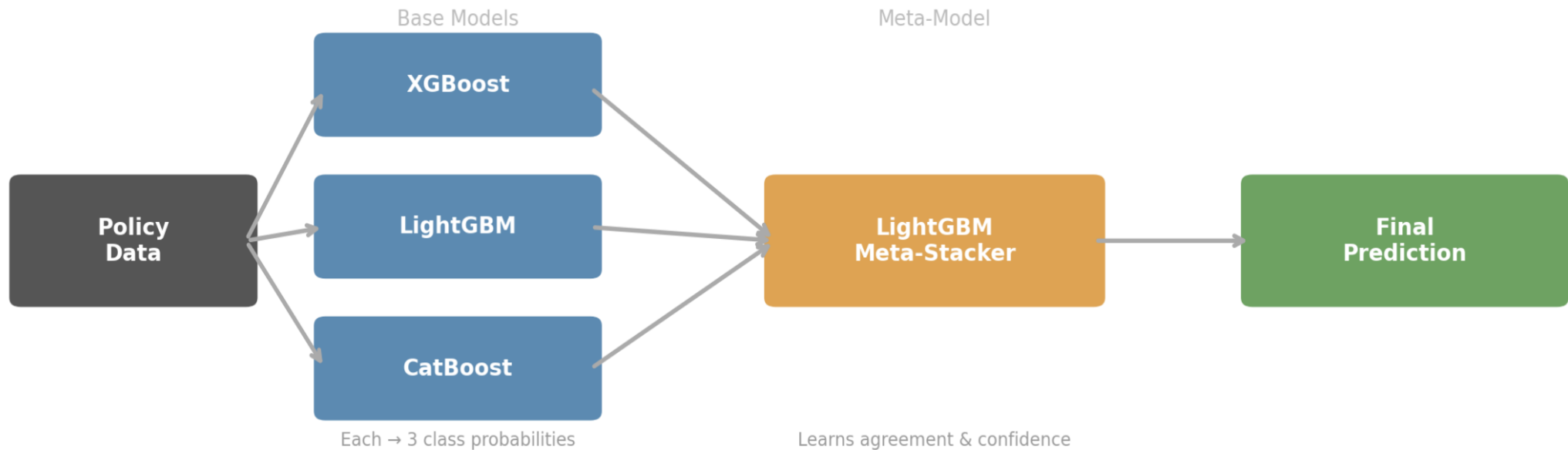
Model Ready Signals

- Customer risk
- Channel risk
- Geographic risk
- Policy risk
- Price or tenure risk
- Missing-value signal

Methods Considered

Method	Strength	Decision
Logistic Regression	Simple, interpretable baseline	Useful for comparison, but too limited for nonlinear patterns
Random Forest	Captures nonlinear relationships and interactions	Considered, but less efficient and not as strong as boosting
Gradient Boosting Models	Strong for tabular data with mixed variable types	Main modeling direction
Blending	Averages probabilities from multiple models	Helped slightly, but fixed weights were limited
Stacking	Learns how to combine model probabilities	Final approach

Final Model: Stacked Ensemble



The final model does not average predictions.

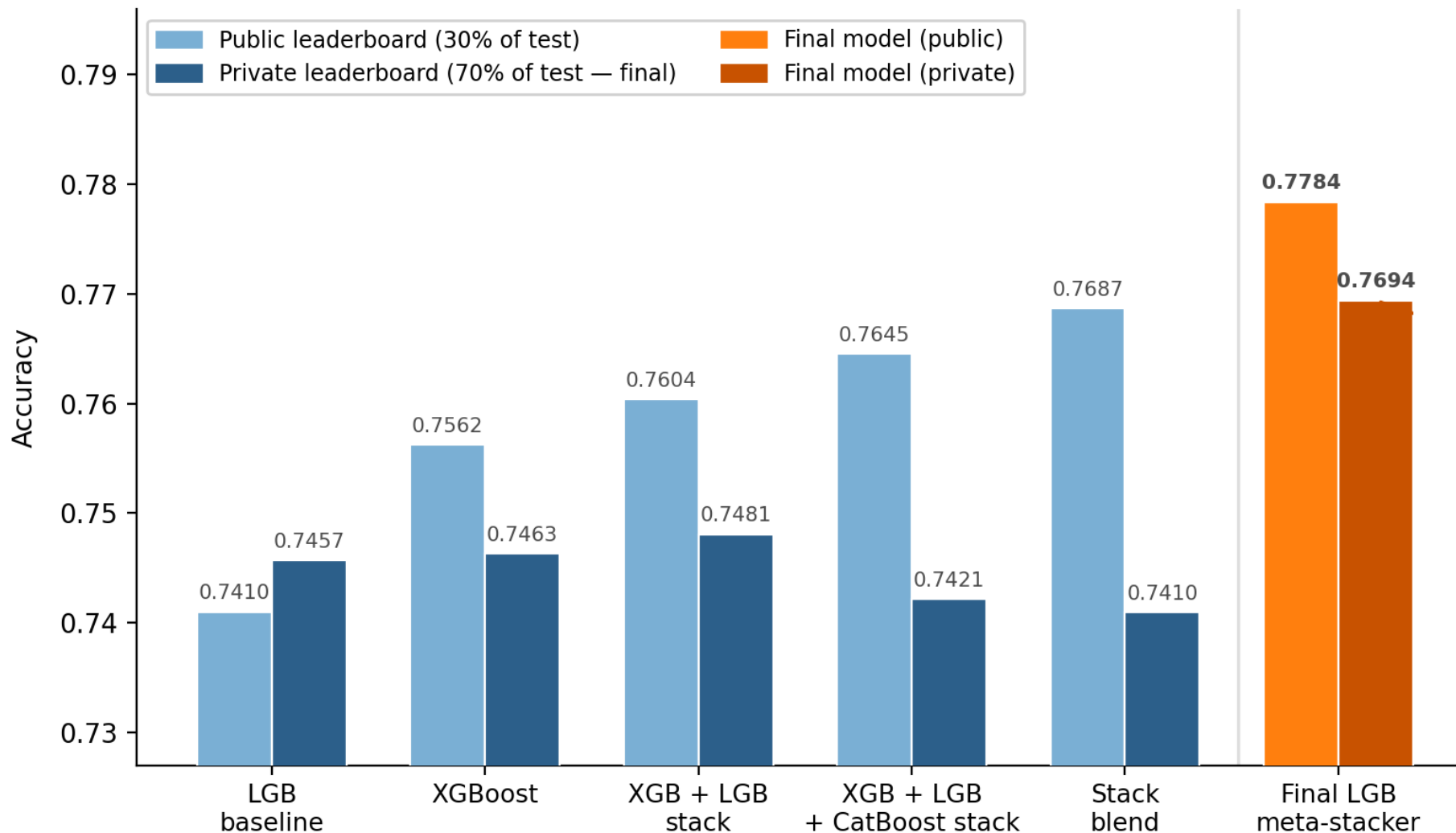
It learns from:

- model confidence
- agreement/disagreement
- probability patterns

Then LightGBM makes the final class prediction.

Results

Model development — public vs. private score by stage



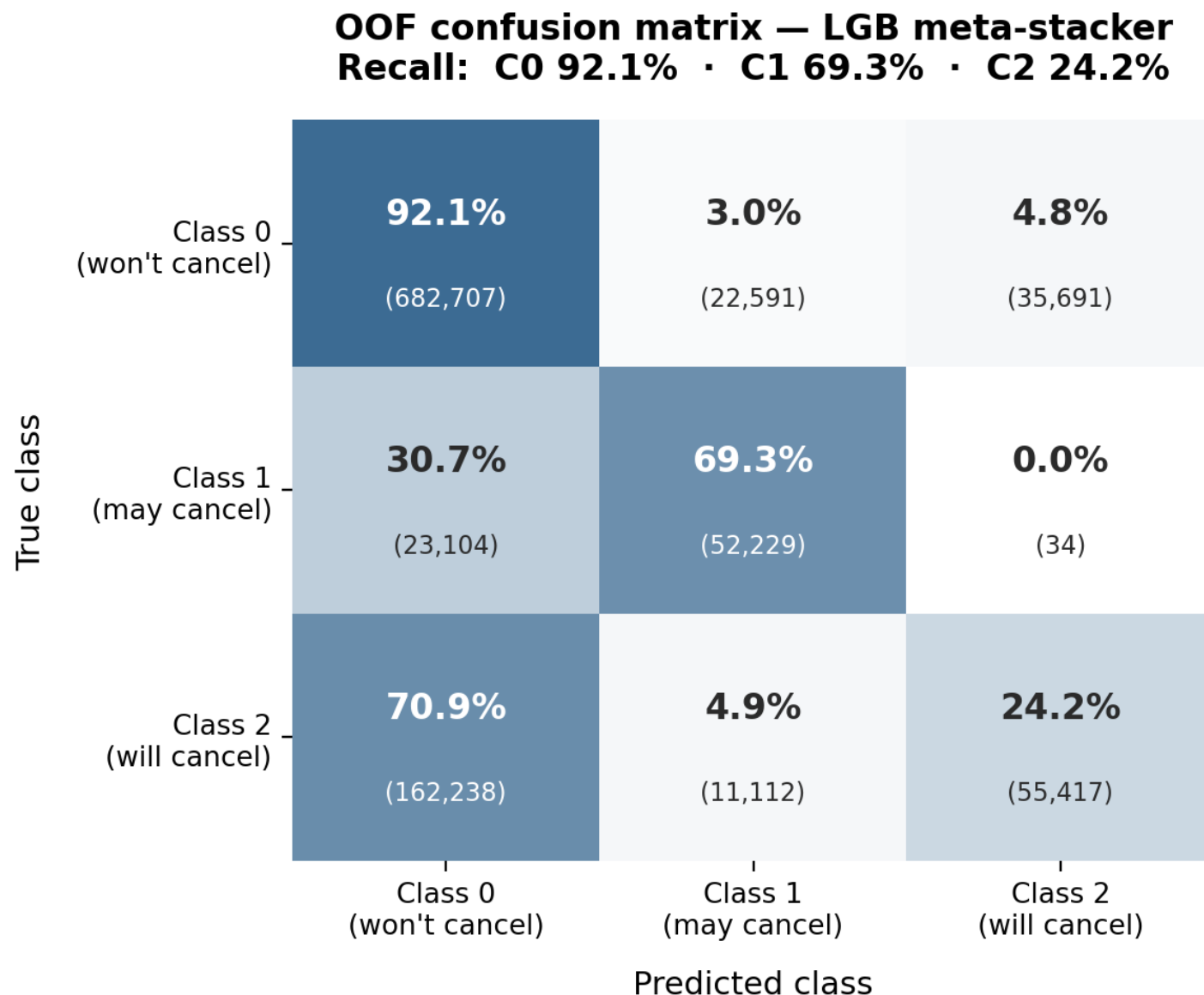
Results

What this shows

- OOF validation results, not hidden test labels
- Class 0 was easiest to identify
- Class 1 recall was strong for a rare class
- Class 2 was often confused with Class 0

Takeaway:

Accuracy was strong overall, but Class 2 recall is a future improvement area.



How Travelers Could Use the Model

Predicted Class	Meaning	Recommended Action
Class 0	Likely to renew	Normal renewal process
Class 1	May cancel but can be convinced	Priority retention outreach for discounts, coverage review, or agent follow up
Class 2	Likely to cancel	Stronger intervention / pricing or coverage review



Additional Variables That Could Help

1. Renewal premium change

- Shows if the customer had a sharp price increase.

2. Payment history / late payments

- Shows financial stress or disengagement.

3. Service calls / complaints

- Shows dissatisfaction before cancellation.

4. Prior cancellation history

- Shows whether the customer has a pattern of leaving.

5. Claim amount and severity

- Separates small claims from major claim experiences.

6. Policy changes near renewal

- Shows if the customer is reconsidering coverage.

7. Bundling / multi-policy status

- Shows customer stickiness and loyalty.

8. Competitor quotes / market context

- Shows outside pressure to switch.

Conclusion

Limitations

- The model is predictive, not causal
- Accuracy was the competition metric, but class-level recall varied
- Class 2 recall was low in OOF validation
- Important behavior and price-change variables were not included

Conclusion

- Built a stacked ensemble for multiclass policy cancellation prediction
- Final model used XGBoost, LightGBM, and CatBoost probabilities
- LightGBM meta-stacker learned confidence and disagreement patterns
- Final private score: 0.76941, 4th place overall

Thank You